

Mini Project: Design and Develop Custom Application (Mini Project) using Salesforce Cloud

Title

Design and Develop Custom Application (Mini Project) using Salesforce Cloud

Aim

To design and develop a custom Lightning Application using Salesforce Cloud Platform by creating a custom object-based application named **Library Management App**, adding custom object **Books**, creating custom fields, managing records, and performing CRUD operations using Salesforce Trailhead Playground. Adapted from your earlier report structure .

Problem Statement

To understand the practical implementation of cloud computing using Salesforce by designing a custom Lightning Application. The project involves creating a Salesforce Trailhead Playground, creating a custom object named **Book**, configuring custom fields such as Author Name, Price, Publisher, Available Copies, creating a custom tab, building a Lightning Application using App Manager, assigning profiles, launching the application, and managing records using CRUD operations on the cloud platform.

Objectives

- To understand the Salesforce cloud platform and its architecture
 - To create and use Trailhead Playground for development
 - To design and configure a custom Lightning Application
 - To understand the use of App Manager in Salesforce
 - To create and manage Custom Objects and Custom Fields
 - To create and verify records inside the custom application
 - To perform CRUD operations using Salesforce
 - To understand cloud-based application development without coding
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Prerequisites

Before performing this mini project, the following are required:

- Basic knowledge of Cloud Computing
- Salesforce Developer Account through Trailhead
- Trailhead Playground environment

Theory

Salesforce Cloud

Salesforce is a cloud-based Customer Relationship Management (CRM) platform that helps organizations manage customer information, sales, services, and business applications over the internet without installing software on local machines.

It provides a complete cloud environment where applications can be created, managed, and accessed from anywhere.

Trailhead Playground

Trailhead Playground is a practice environment provided by Salesforce where users can safely learn, test, and build applications without affecting real business data.

It acts as a personal Salesforce developer environment where students can perform experiments and mini projects.

Lightning Experience

Lightning Experience is the modern graphical user interface of Salesforce that provides a better user experience and supports app development using drag-and-drop tools and configuration-based development.

It simplifies application creation and improves productivity.

App Manager

App Manager is used to create and manage Lightning Applications in Salesforce.

It helps in:

- Creating new applications
 - Customizing navigation menus
 - Adding standard and custom objects
 - Managing user profiles and permissions
 - Configuring app branding and accessibility
-

Custom Objects in Salesforce

Custom Objects are user-defined database tables created according to project requirements.

Each object stores a specific type of information.

Custom Object Used

Book

Stores library-related book information.

Example Fields:

- Book Name
- Author Name
- Publisher
- Price
- Length
- Available Copies

Lightning Application

A Lightning Application is a customized workspace in Salesforce where users can access specific objects and features based on business requirements.

In this mini project, a custom application named **Library Management App** is created using App Manager by adding a custom object called **Books**.

This demonstrates practical cloud application development using no-code configuration.

Implementation with Outputs

Step 1: Create Trailhead Playground

First, Salesforce Trailhead account was created and login was completed.

Navigation:

Trailhead → Profile → Hands-On Orgs → Create Playground

This provides a cloud-based development environment for performing the project.

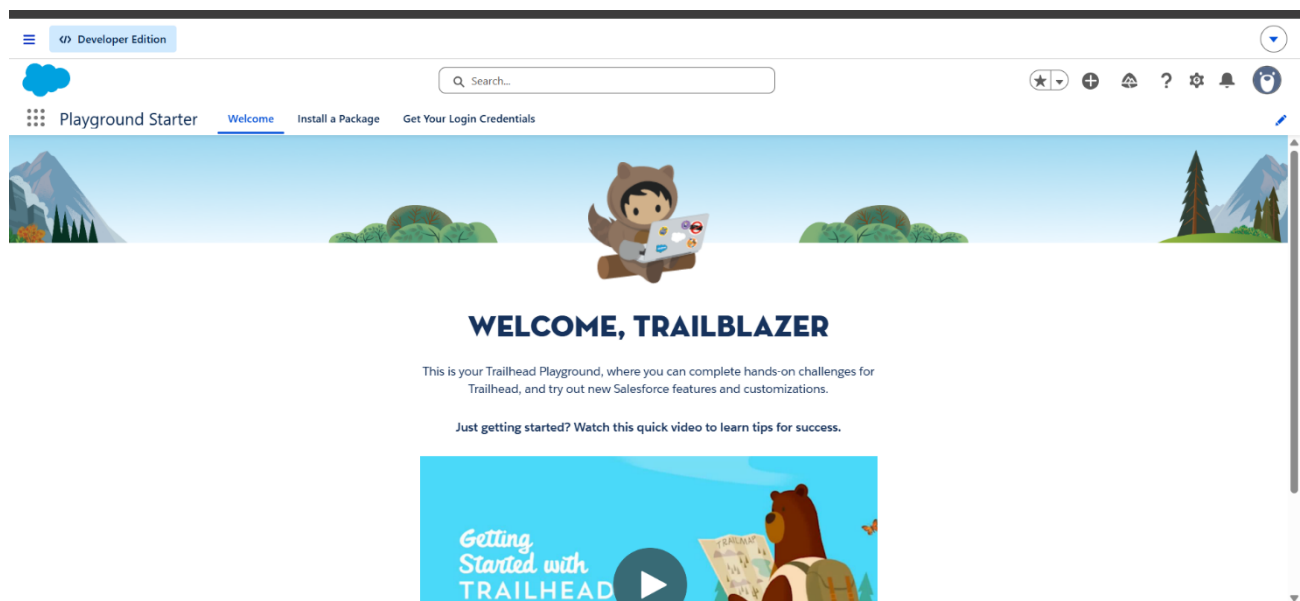


Fig 1: Trailhead Playground Dashboard

Step 2: Open Playground

The created Playground was launched by clicking the Launch button.

This opens the Salesforce development environment where the custom application is created.

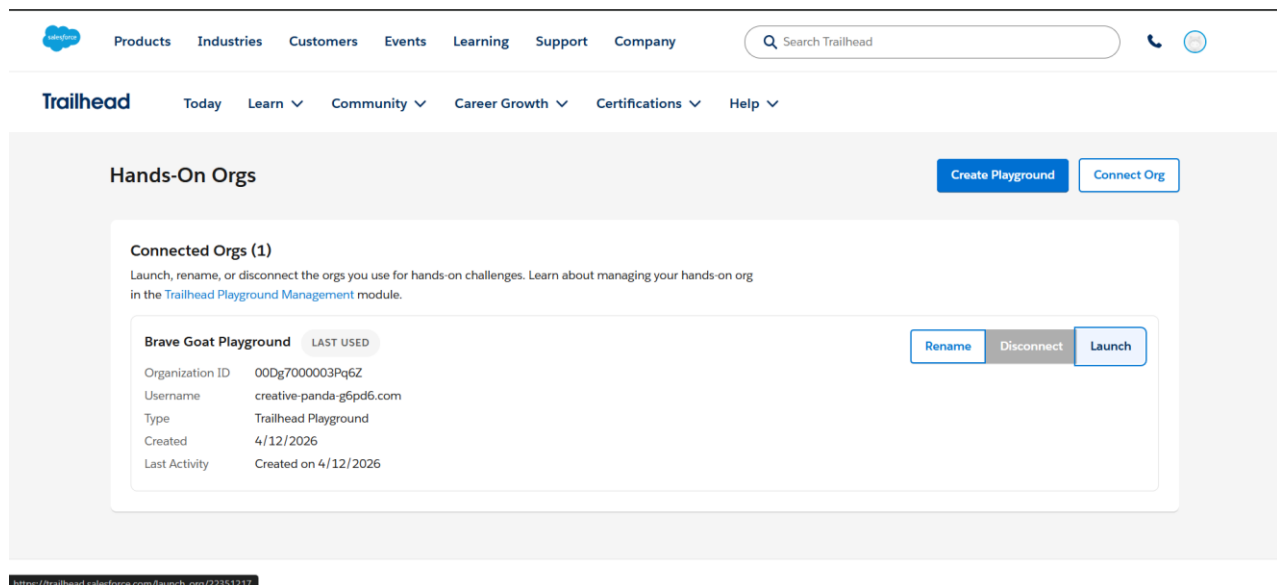


Fig 2: Playground Launch Window

Step 3: Open Setup

Inside the Playground, Setup was opened using the Gear Icon.

Navigation:

⚙️ → Setup

This opens the Salesforce administration panel.

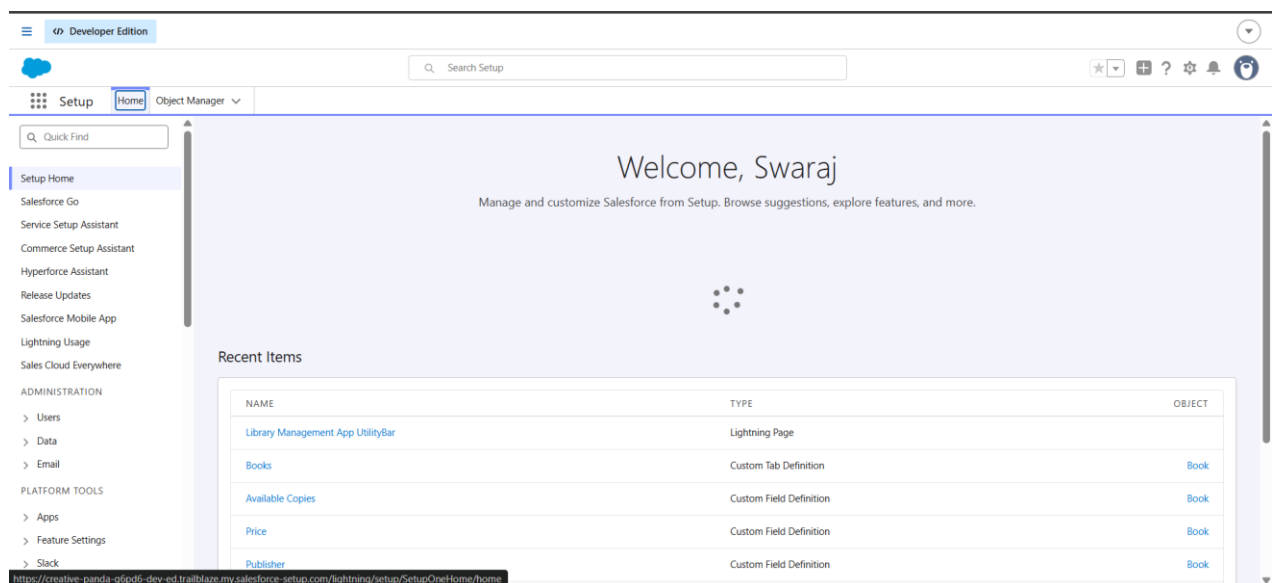


Fig 3: Setup Window

Step 4: Open Object Manager

In Setup, Object Manager was searched and opened.

Navigation:

Setup → Object Manager

This section is used for creating custom objects.

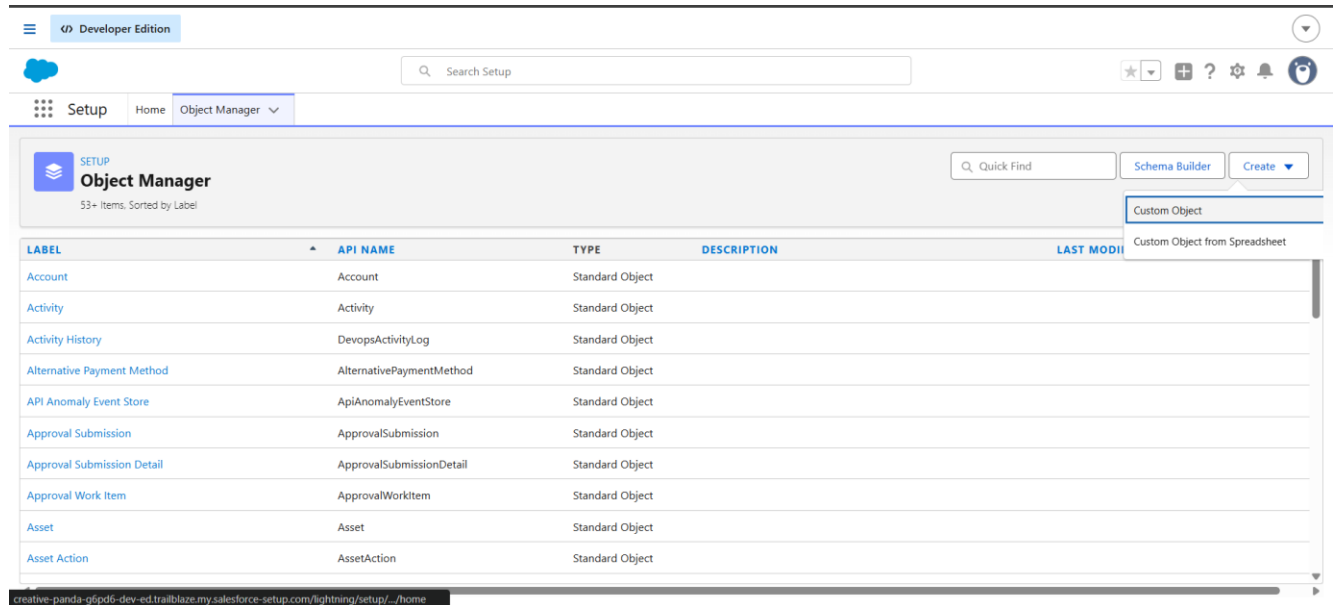


Fig 4: Object Manager Page

Step 5: Create Custom Object

A new Custom Object was created.

Details entered:

- Label: Book
- Plural Label: Books
- Object Name: Book
- Record Name: Book Name
- Data Type: Text

This creates the base structure of the project.

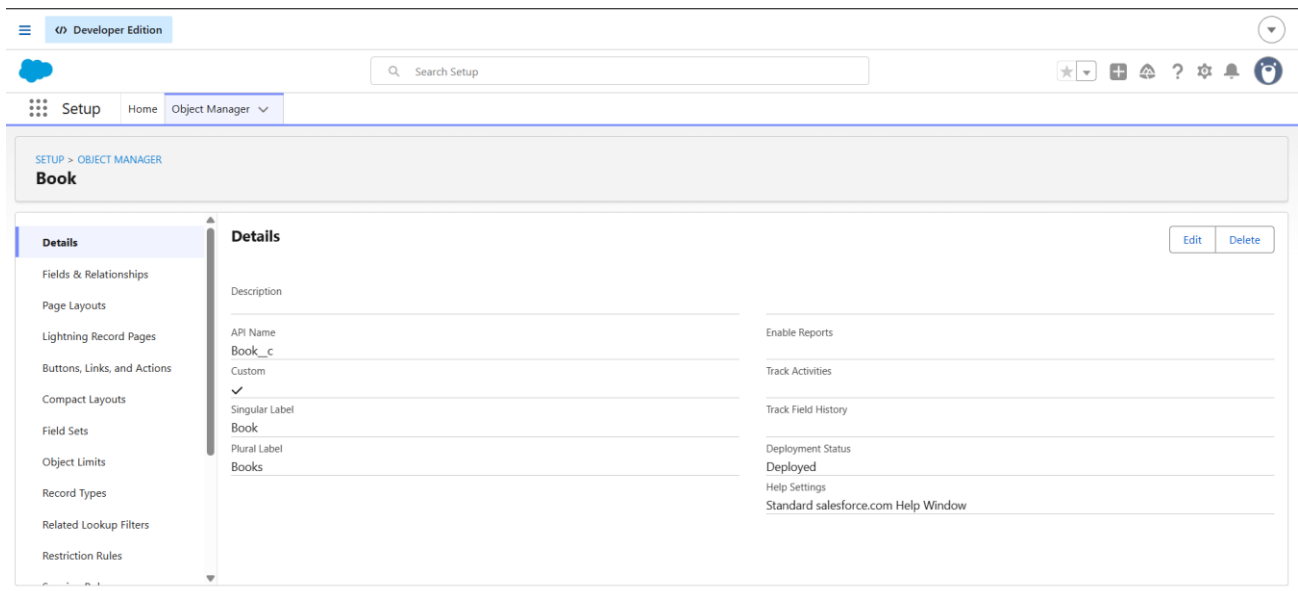


Fig 5: Custom Object Creation

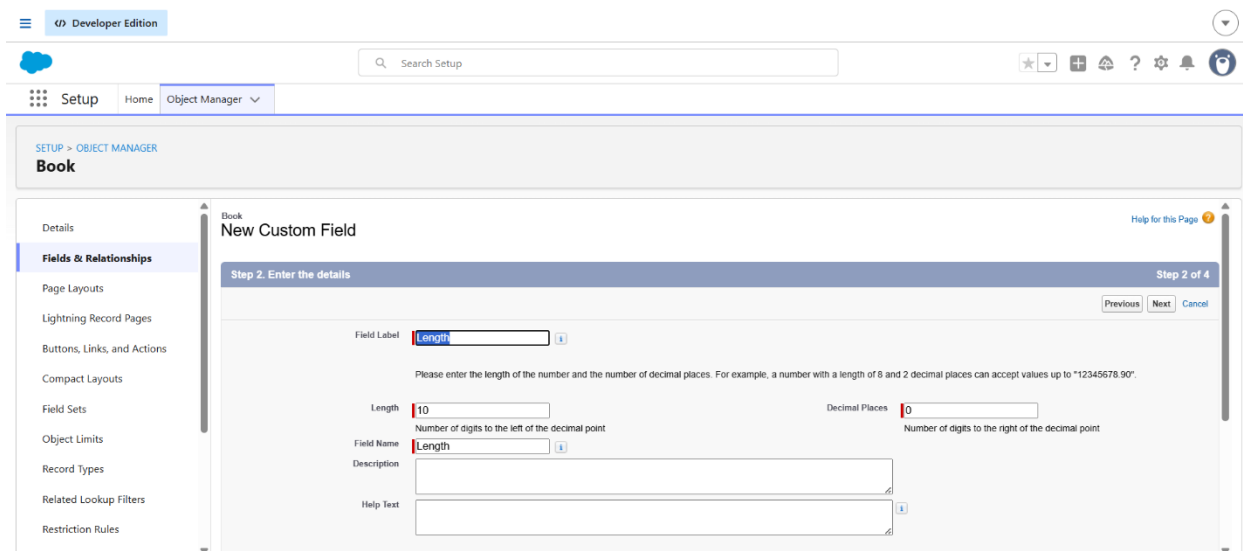
Step 6: Create Custom Fields

Custom fields were created inside the Book object.

Fields Added:

- Author Name
- Publisher
- Price
- Length
- Available Copies

These fields store complete book details.



Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Book

Details

Fields & Relationships
9 Items. Sorted by Field Label

Q Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Author Name	Author_Name__c	Text(50)		
Available Copies	Available_Copies__c	Number(10, 0)		
Book Name	Name	Text(80)		✓
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Length	Length__c	Number(10, 0)		
Owner	OwnerId	Lookup(User, Group)		✓
Price	Price__c	Currency(18, 0)		
Publisher	Publisher__c	Text(50)		

Fig 6: Custom Fields Creation

Step 7: Create Custom Tab

A custom tab was created for the Book object.

Navigation:

Setup → Tabs → Custom Object Tabs → New

This allows the Book object to appear in the application menu.

Developer Edition

Setup Home Object Manager

Q Quick Find

SETUP

Tabs

New Custom Object Tab

Step 1. Enter the Details Step 1 of 3

Choose the custom object for this new custom tab. Fill in other details.

New Custom Object Tab Required information

Select an existing custom object or create a new custom object now.

Object: --None--

Tab Style: --None--

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link: --None--

Enter a short description.

Description:

Next Cancel

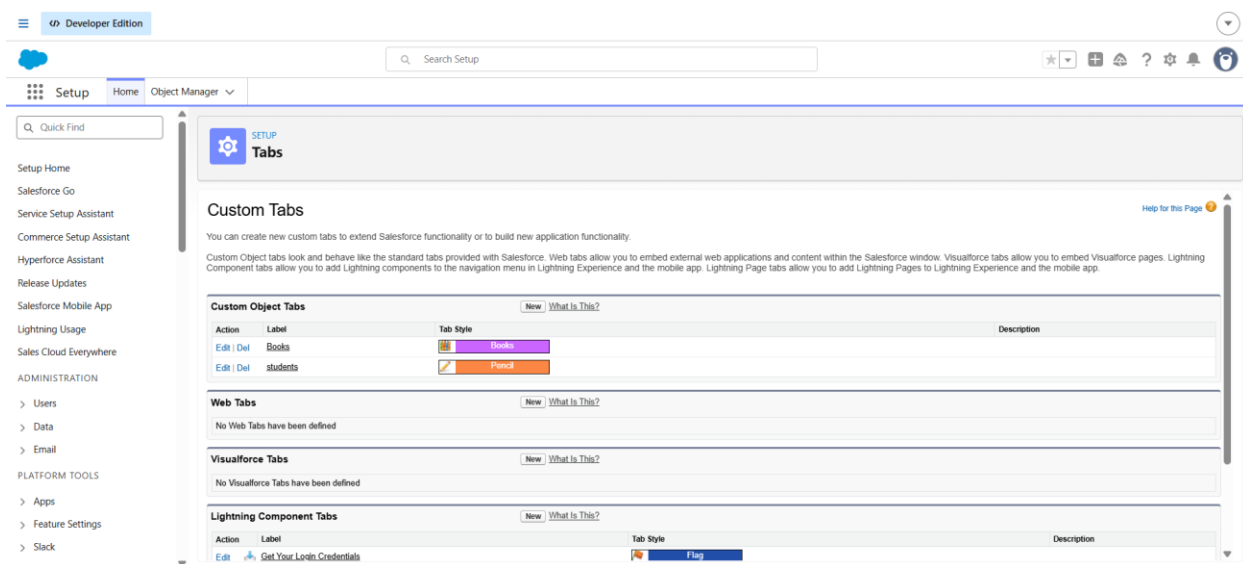


Fig 7: Custom Tab Creation

Step 8: Create New Lightning App

A new Lightning Application was created.

Details entered:

- App Name: Library Management App

This creates the main application structure.

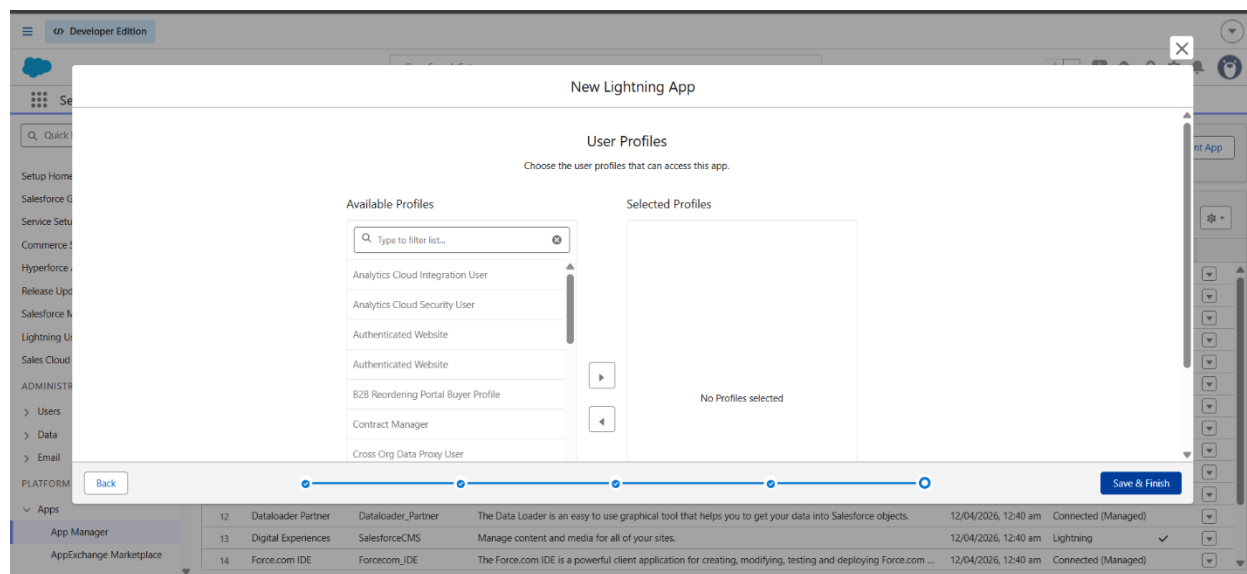


Fig 8: New Lightning App Creation

Step 9: Add Navigation Items

The custom object was added to application navigation.

Added Items:

- Books

This makes the object directly accessible from the app menu.

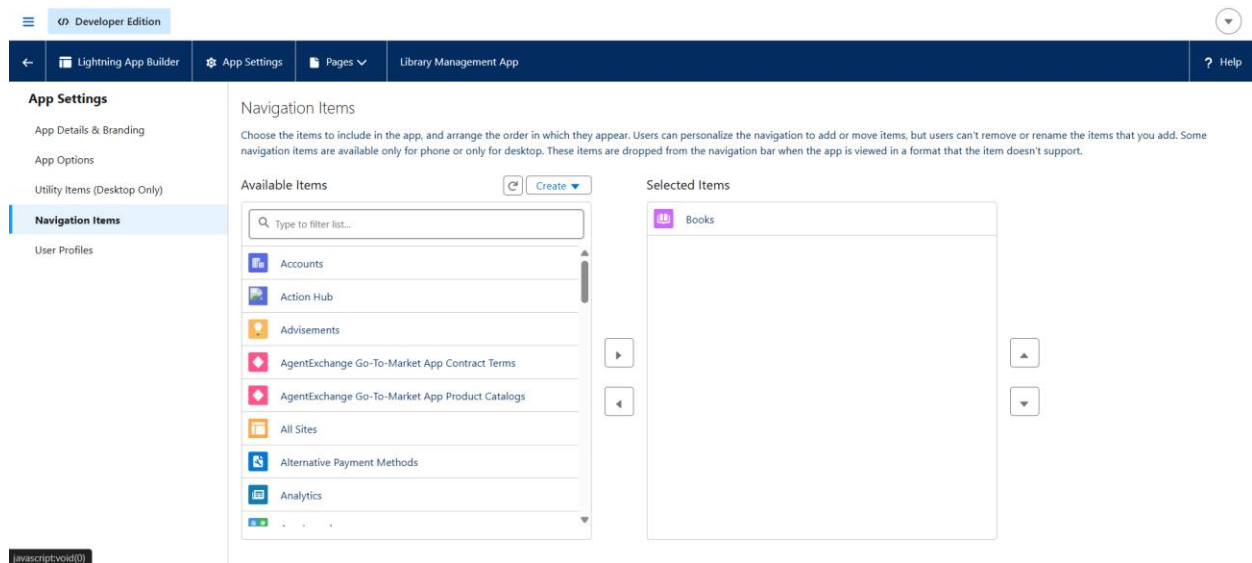


Fig 9: Adding Navigation Items

Step 10: Assign User Profiles

User profiles were assigned for application access.

Selected Profile:

- System Administrator

This gives permission for managing and accessing the application.

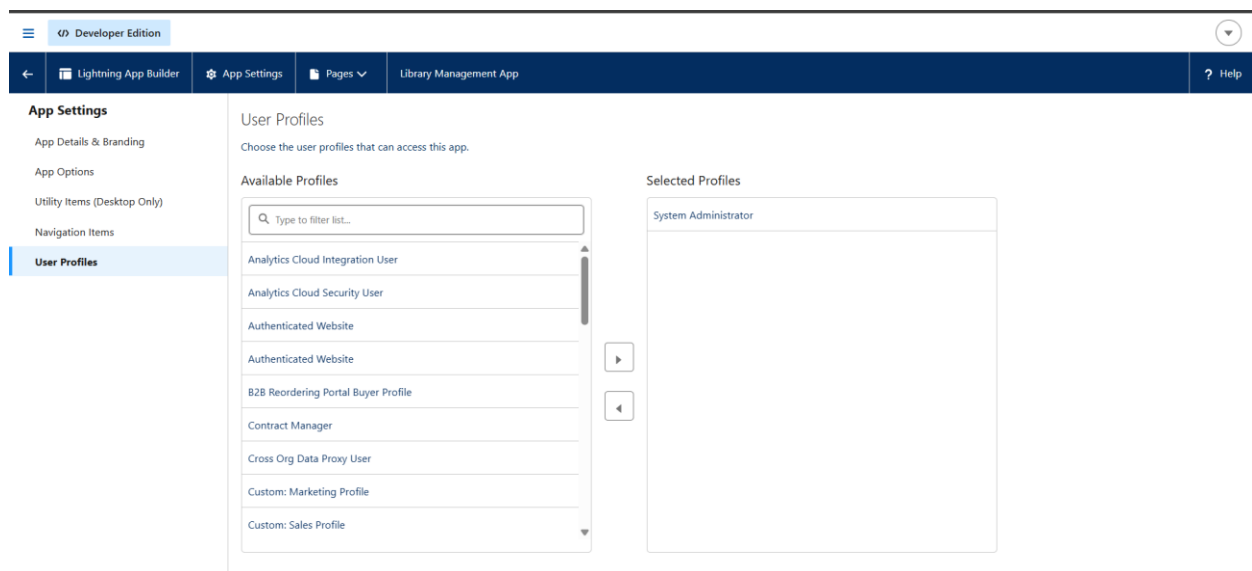


Fig 10: Profile Assignment

Step 11: Save and Launch Application

The application setup was completed using Save & Finish.

Then the app was launched using:

Navigation:

App Launcher → Library Management App

The custom Lightning Application opened successfully.

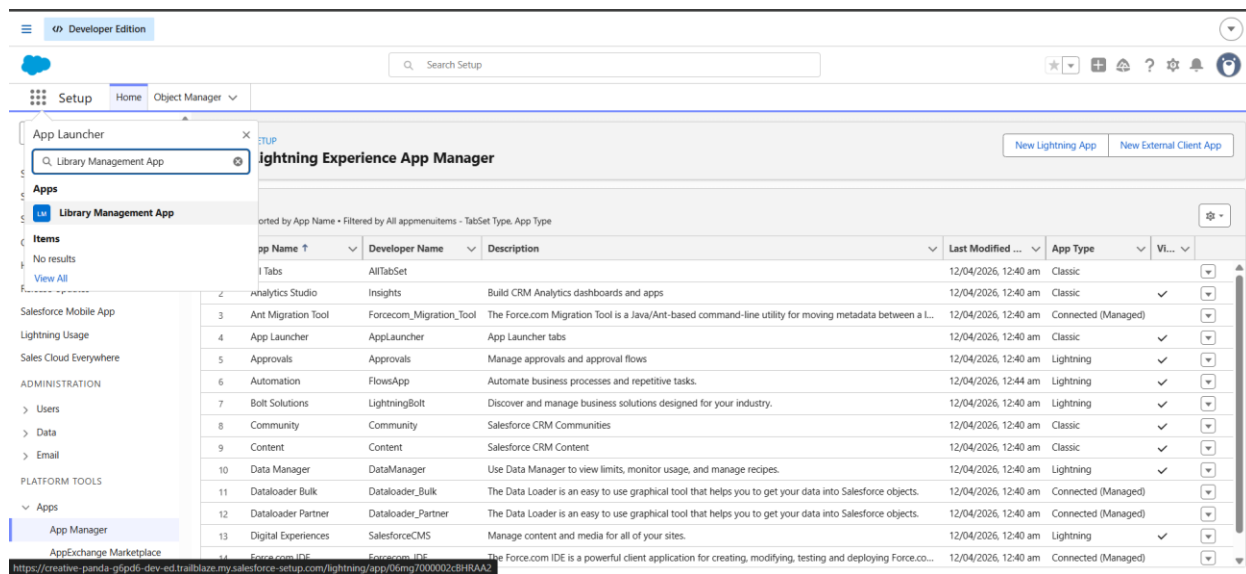


Fig 11: Launching Library Management App

Step 12: Create Book Records

Inside the Books object, new records were created.

Example Records:

Record 1

- Book Name: Rich Dad Poor Dad
- Author Name: Robert Kiyosaki
- Publisher: Plata Publishing
- Price: 499
- Length: 336
- Available Copies: 10

Record 2

- Book Name: Atomic Habits
- Author Name: James Clear
- Publisher: Avery
- Price: 599
- Length: 320
- Available Copies: 8

This demonstrates record creation inside the application.

Developer Edition

Library Management App

Books

Recently Viewed

0 items • Updated a few seconds ago

Nothing to see here

New Book

* = Required Information

Information

* Book Name Rich Dad Poor Dad

Length 336

Author Name Robert Kiyosaki

Publisher Plata Publishing

Price ₹499

Available Copies 10

Owner Swaraj Ambawade

Cancel Save & New Save

Developer Edition

Library Management App

Books

Recently Viewed

0 items • Updated a few seconds ago

Nothing to see here

New Book

* = Required Information

Information

* Book Name Rich Dad Poor Dad

Length 336

Author Name Robert Kiyosaki

Publisher Plata Publishing

Price ₹499

Available Copies 10

Owner Swaraj Ambawade

Cancel Save & New Save

Fig 12: Creating Book Records

Step 13: Read Operation

The saved records were opened and viewed successfully.

This confirmed successful data storage in the cloud object.

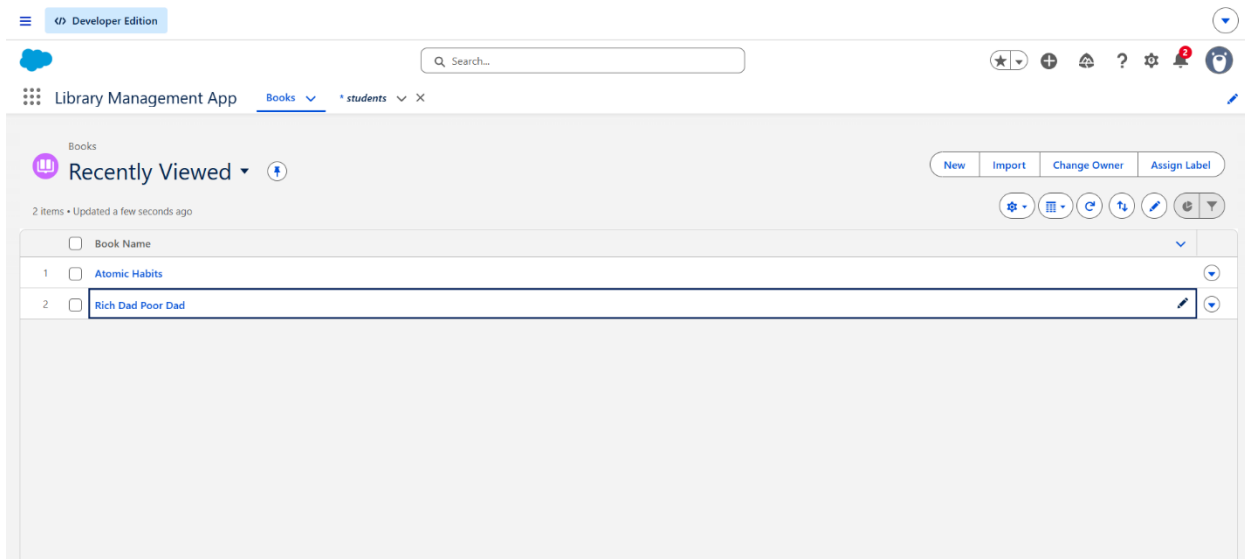


Fig 13: Read Operation

Output

The custom Lightning Application named **Library Management App** was successfully created using Salesforce Cloud Platform.

A custom object named **Book** was created along with multiple custom fields such as Author Name, Price, Publisher, Length, and Available Copies.

The Books object was added to the application navigation, records were created successfully, and CRUD operations (Create, Read, Update, Delete) were performed successfully.

The application was launched successfully and all configured features worked correctly.

Conclusion

The mini project on designing and developing a custom application using Salesforce Cloud was completed successfully.

A custom Lightning Application named **Library Management App** was created using App Manager. A custom object named **Book** was developed with required fields, user profiles were assigned, and records were managed successfully using CRUD operations.

This project provided practical understanding of Salesforce cloud platform, CRM application development, custom object creation, and no-code/low-code cloud-based application design.

It also helped in understanding how enterprise applications are built and managed using cloud services.